Statistical Signal Processing
Lecture 12:
Adaptive filters and the RLS algorithm

Literature and Problems: See course schedule

Agenda (preliminary):

Review

- **Generic model for adaptive filtering**
- **SLIDE: Optimal adaptive filter – Wiener Hopf equation**

RLS algorithm vs. LMS algorithm approach to iteration towards optimal filter
RLS algorithm vs. Kalman filter (brief)

First step: from ensemble mean to time average version of autocorrelation matrix and crosscorrelation vector

Second step: limitation of RLS algorithm "memory" by introducing forgetting factor

- **Backward exponential window and equivalent "memory" (backward dependency)**
- **Modified cost function**

The standard RLS algorithm

- **Recursive solution of Wiener Hopf equation for optimal filter**
- **Recursive equations for auto- and cross-correlation estimates**
- **Recursive equation for filter coefficients**
- **Recursive equation for autocorrelation matrix inverse using the Sherman-Morrison inversion formula**

SLIDE: The standard RLS algorithm and its performance

- **Convergence and unbiased estimation**
- **Convergence speed**
- **Stationary and excess mean square error**

SLIDE: Applications of adaptive filtering
DEMO: Adaptive System Identification
DEMO: Adaptive Linear Prediction and Noise Reduction